Multiscale Variable-Horizon Bond-Based Peridynamics

Graded SALOME Meshes (MED / UNV) | $\delta_i = m \cdot \Delta x_i$ $(m = 3)$ | Symmetric KD-Tree & Dual-Horizon PD

1. Mathematical Formulation & Multiscale Implementation

- **Nodal Volume & Multiscale Horizon from SALOME (MED / UNV):**

  Each element e of measure V_e distributes $V_e/n_{v,e}$ to its vertices. The local node spacing and horizon are:
  $\Delta x_i = V_i^{1/3}$ (3D), $\Delta x_i = A_i^{1/2}$ (2D), $\Delta x_i = L_i$ (1D), $\delta_i = m \cdot \Delta x_i$ $(m = 3)$.

- **KD-Tree Neighbor Search & Symmetric Family Condition:**

  Candidate neighbors are queried with a KD-tree (scipy.spatial.cKDTree). Each undirected bond (i, j) with $i < j$ is stored once and kept if and only if $|\boldsymbol{\xi}_{ij}| = \|\mathbf{x}_j - \mathbf{x}_i\| \le \min(\delta_i, \delta_j)$.
  One-sided bonds $(\min(\delta_i, \delta_j) < |\boldsymbol{\xi}_{ij}| \le \max(\delta_i, \delta_j))$ are strictly excluded so families are symmetric $(j \in \mathcal{H}_i \Leftrightarrow i \in \mathcal{H}_j)$.

- **Symmetric Micromodulus & Single-Loop Pairwise Force Assembly (Exact Newton's 3rd Law):**

  With $\delta_{ij} = \min(\delta_i, \delta_j)$, the bond micromodulus $c_{ij} = c(\delta_{ij}) = c_{ji}$ is strictly symmetric:
  $c(\delta_{ij}) = \frac{2E}{\delta_{ij}^2}$ (1D), $c(\delta_{ij}) = \frac{9E}{\pi\delta_{ij}^3}$ (2D plane stress), $c(\delta_{ij}) = \frac{12E}{\pi\delta_{ij}^4}$ (3D).
  In a single loop over stored bonds $b = (i, j)$, $+\mathbf{F}_{ij}$ is added to i and $-\mathbf{F}_{ij}$ to j, with $\mathbf{F}_{ij} = g_{ij} c_{ij} s_{ij} \frac{\boldsymbol{\xi}_{ij}}{|\boldsymbol{\xi}_{ij}|} V_i V_j$.

- **Stiffness Correction $g_{ij} = \frac{1}{2}(\lambda_i + \lambda_j)$ & Transition-Zone Ghost-Force Elimination:**
  Comparing discrete energy density $W_i^{\mathrm{PD},0} = \frac{1}{4}\varepsilon_0^2 \sum_{j \in \mathcal{H}_i} c_{ij} |\boldsymbol{\xi}_{ij}| V_j$ under uniform isotropic strain $\boldsymbol{\varepsilon} = \varepsilon_0 \mathbf{I}$ against continuum elasticity W^{cont} yields $\lambda_i = W^{\mathrm{cont}} / W_i^{\mathrm{PD},0}$ and $g_{ij}^{(0)} = \frac{1}{2}(\lambda_i + \lambda_j)$.
  A minimum-L^2-norm projection $g_{ij} = g_{ij}^{(0)} + \Delta g_{ij}$ enforces vanishing 3rd-order bond moments $\sum_{j \in \mathcal{H}_i} g_{ij} c_{ij} (\mathbf{n}_{ij}^{\otimes 3}) V_j = \mathbf{0}$ at interior/transition nodes, eliminating transition-zone ghost forces.

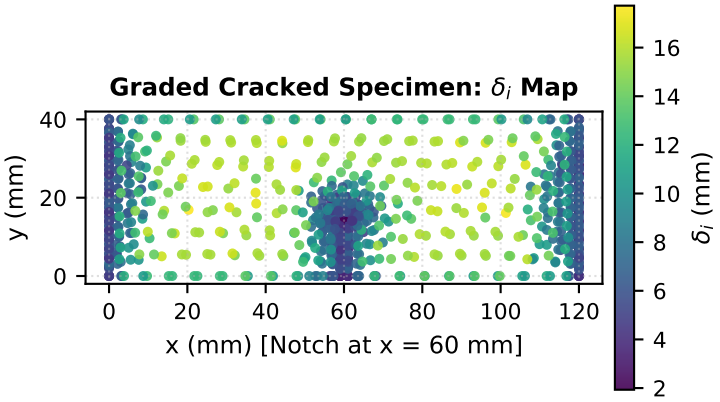
- **Dual-Horizon Peridynamics Mode (Optional Comparison):**

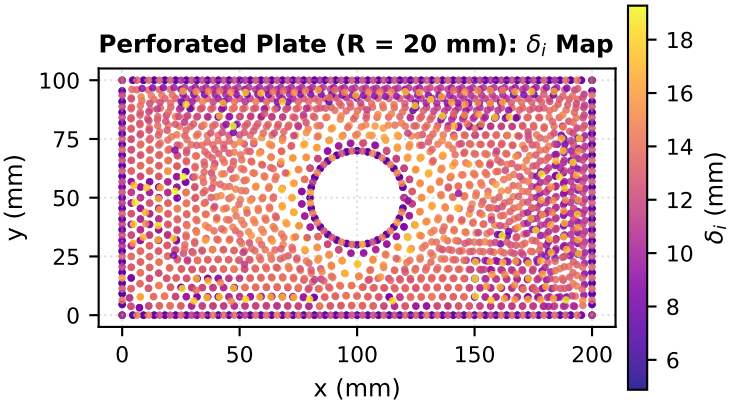
  In mode = dual_horizon (Ren et al., 2016), bonds up to $\max(\delta_i, \delta_j)$ are retained with effective weight $\tilde{c}_{ij}^{\mathrm{DH}} = \frac{1}{2} c(\delta_i) \mathbb{1}(|\boldsymbol{\xi}_{ij}| \le \delta_i) + \frac{1}{2} c(\delta_j) \mathbb{1}(|\boldsymbol{\xi}_{ij}| \le \delta_j)$.

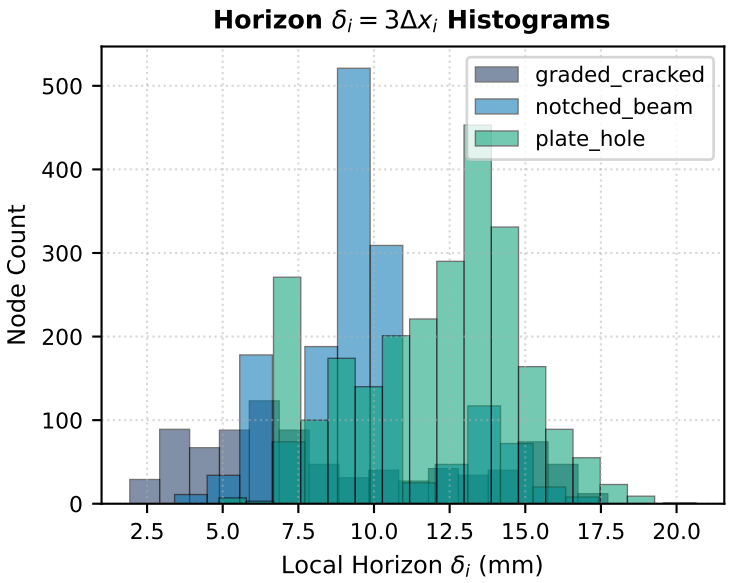
2. Executive Summary of Automated Verification Suite (tests/test_multiscale_pd.py)

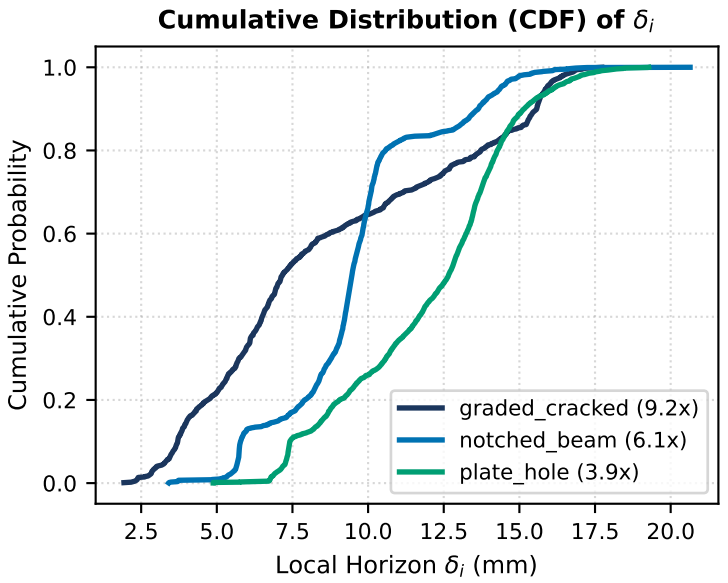
Test Case	Requirement Verified	Quantitative Metric	Status
Test 1: KD-Tree & Symmetric Families	Single bond storage (i < j), |xi| <= min(delta_i, delta_j), exclude one-sided bonds	2D Graded: 6,050 sym. bonds, 530 one-sided excluded 3D SALOME: 10,677 sym. bonds, 17,437 excluded Exact symmetry: j in H_i <=> i in H_j	PASS
Test (a): Total Internal Force = 0	Symmetric c_ij = c(delta_ij), +f to i and -f to j in same loop (Newton's 3rd Law)	1D / 2D / 3D & SALOME Graded Meshes: ||sum_i F_i|| / sum_i ||F_i|| = 3.13e-17 (< 1e-14) Verified for linear & nonlinear kinematics	PASS
Test (b): Patch Test & Zero Ghost Force	Zero ghost force under uniform strain, including across the fine/coarse transition zone	Max Transition-Zone |f_ghost| (Static BVP Err): 1D: 2.80e-13 (4.94e-15) | 2D: 8.01e-14 (6.55e-16) 3D: 1.36e-14 (3.08e-16)	PASS
Test (c): 1D Wave Fine/Coarse Interface	Gaussian pulse crossing dx_fine -> 2*dx_fine interface with no spurious reflection	Uncorrected (g=1): 7.54% refl | Energy g0: 3.64% refl Dual-Horizon PD: 1.72% refl, 100.02% trans Multiscale Corrected: 0.30% refl, 100.12% trans	PASS
Test 5: SALOME Mesh (MED / UNV) & delta_i	Read V_i, dx_i, delta_i = 3*dx_i from SALOME .med & .unv and report delta distribution	MED vs. UNV match: max |d_delta| = 8.67e-18 m Graded Cracked: [1.93, 17.72] mm (9.18x ratio) Notched Beam: [3.41, 20.64] mm (6.05x ratio)	PASS

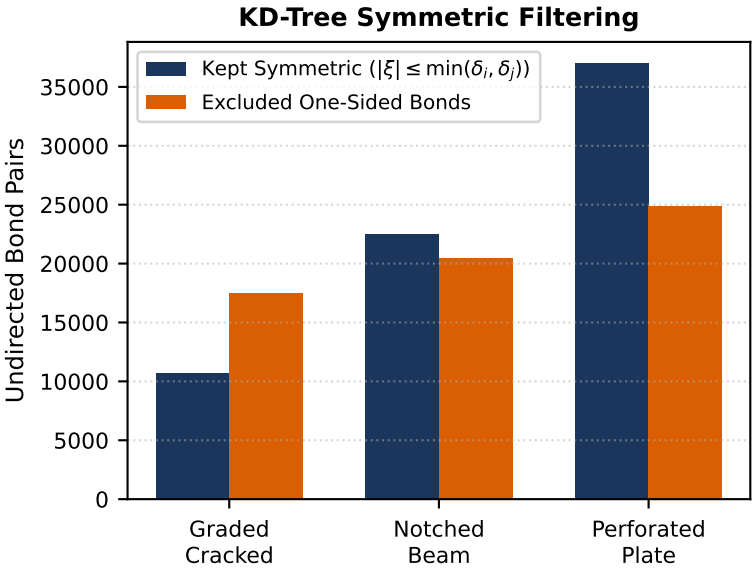
SALOME Graded Mesh Exports (MED & UNV): Node Size & Horizon ($\delta_i = 3\Delta x_i$)







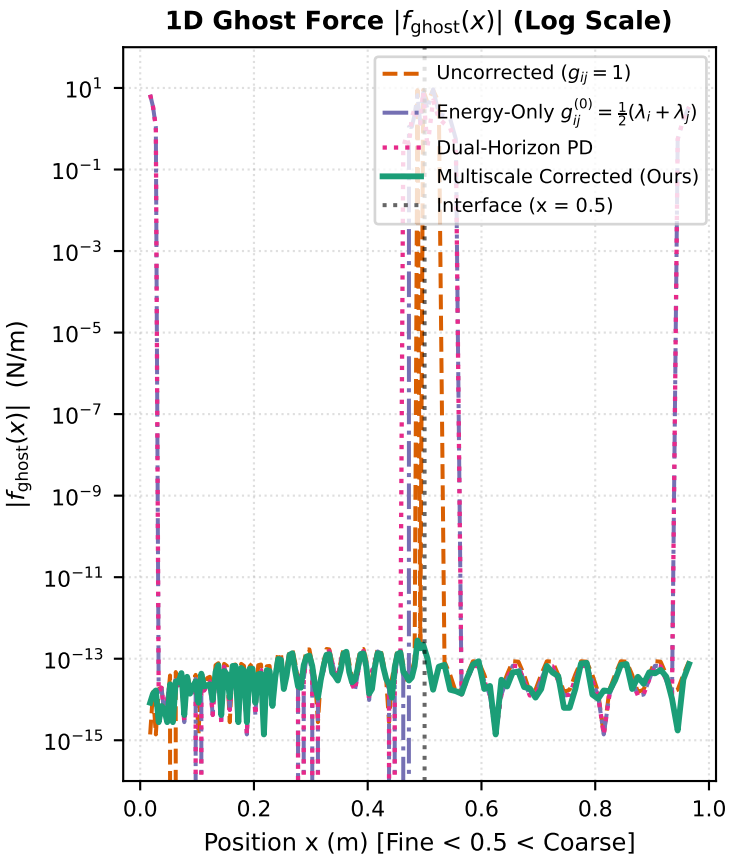


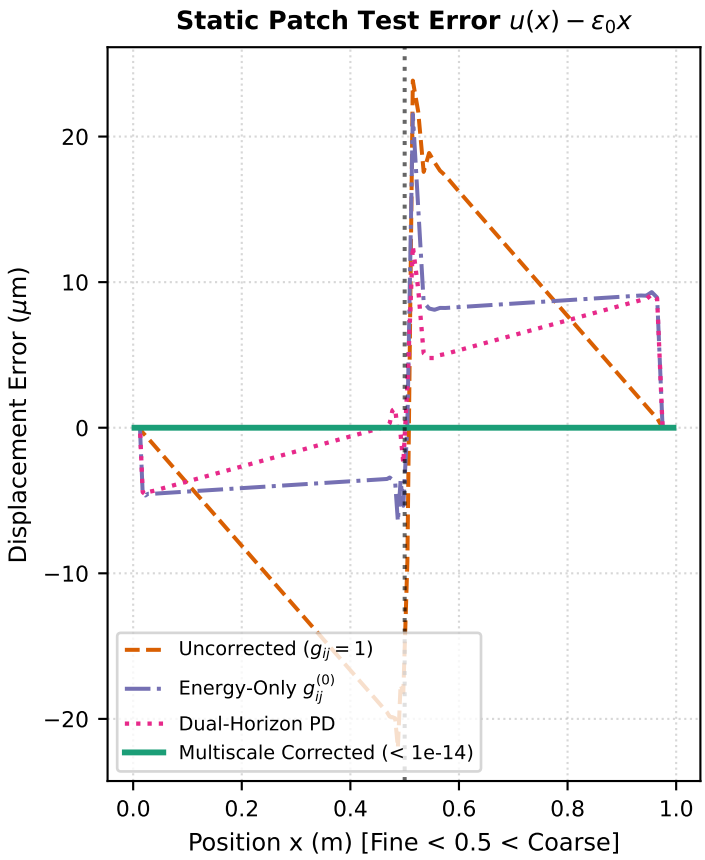


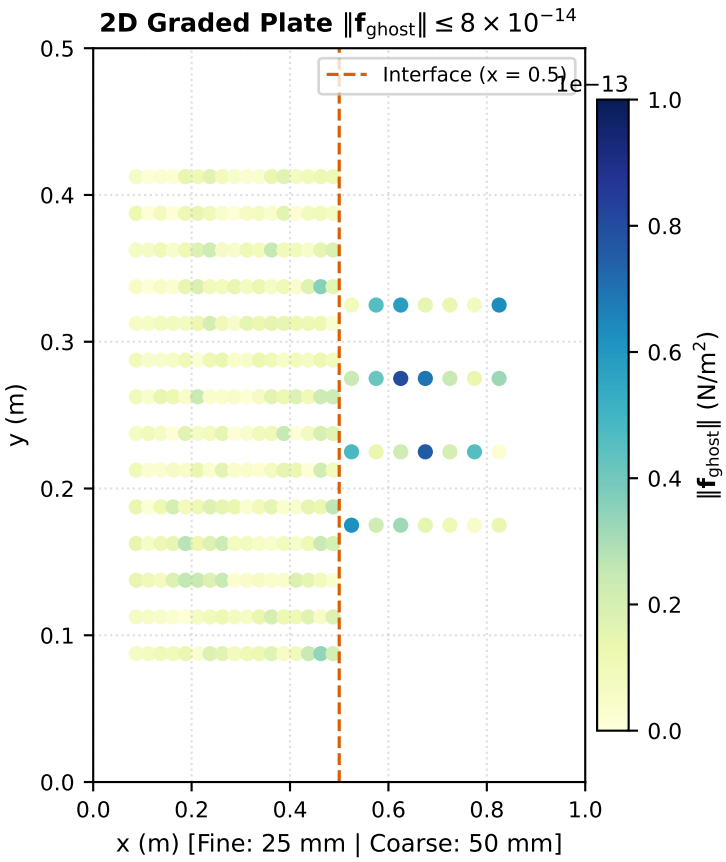
SALOME MED & UNV Statistics ($m = 3$)

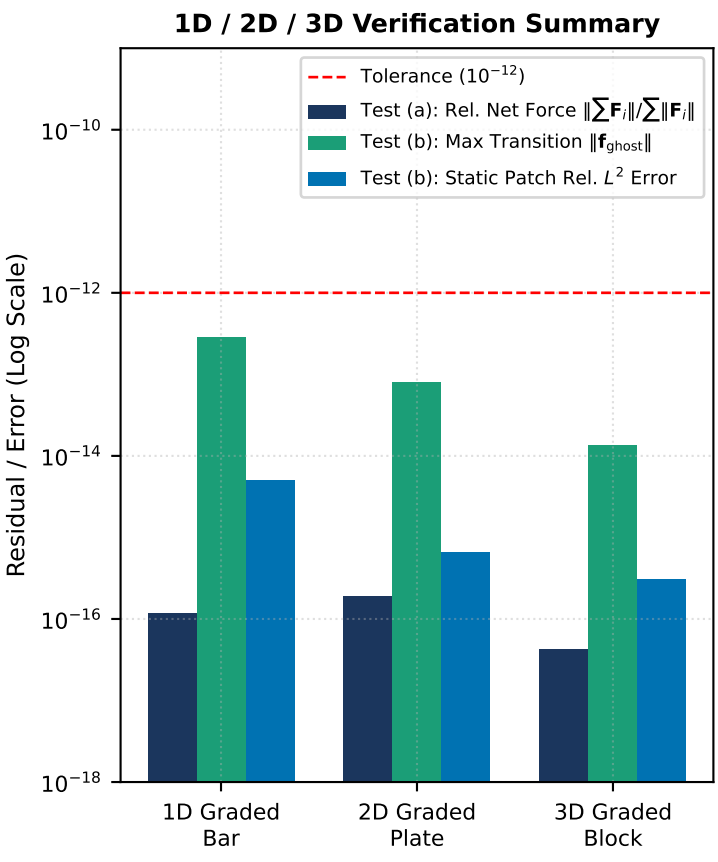
Mesh	N / Tet	dx (mm)	delta (mm)	Mean	Ratio
GradedCrack	878/2609	0.64-5.91	1.93-17.72	8.69	9.18x
NotchedBeam	1605/5188	1.14-6.88	3.41-20.64	9.59	6.05x
PlateHole	2531/7550	1.63-6.42	4.88-19.27	12.00	3.95x

Test (a) Exact Newton's 3rd Law & Test (b) Uniform-Strain Patch Test









Test (c): 1D Wave Packet Crossing Fine/Coarse Interface ($x = 0.50$ m)

